

Client: Sydney Water Corporation
Project: Willoughby Submain Amplification Project
Location: Waters Road, Naremburn, NSW 2065

Project No: 60764076
Logged by: KM
Start Date: 29/06/26
Checked by: CJ
End Date: 02/07/26
Location Meth.: RTK
RL: 62.26 m
Ver. Datum: AHD
Surface: Topsoil

Driller: Terratest
Drill Rig: Comacchio GEO 305
Total Depth: 25.92 m
Hole Diameter: 200-96 mm
Inclination: 90°
Bearing: N/A

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
HA	U	ES: 0.10-0.20 m PP=400, 420, 270 kPa PID=0.3 PPM D: 0.35-0.45 m ES: 0.35-0.45 m PP=350-310, 400 kPa		62.0	0.0		CI	TOPSOIL - Silty CLAY: medium plasticity, brown. FILL - Sandy CLAY: low plasticity, brown; sand, fine to medium grained; with gravel, medium to coarse, angular to subangular, iron stone. Silty Sandy CLAY: medium plasticity, pale grey mottled orange and red-brown; sand, fine grained.	w ≈ PL		TOPSOIL FILL RESIDUAL SOIL
					1.0			SANDSTONE: medium grained, red brown and brown, very thinly cross-bedded at 5-20°, with iron bands up to 10 mm thick. <i>Log continued on next page.</i>	w < PL	VSt	HAWKESBURY SANDSTONE
					61.0						
					2.0						
					60.0						
					3.0						
					59.0						
					4.0						
					58.0						
					5.0						
					57.0						
					6.0						
					56.0						
					7.0						
					55.0						
					8.0						

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Easting: 333075.60
Northing: 6256476.00
Hor. Datum: MGA2020-56

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x IS ₍₅₀₎ A: • D: o UCS: □ SOIL: VL L M H VH EH	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
					0.0							
					62.0							
							<i>Log continued from previous page.</i>					
	1		Is(50) D=1.79 MPa Is(50) A=2.83 MPa C: 1.25-1.45 m UCS=16 MPa		1.0		SANDSTONE: medium grained, red brown and brown, very thinly cross-bedded at 5-20°, with iron bands up to 10 mm thick. from 0.82 to 0.91 m: ironstone band, 90 mm thick	HW MW	100 (87) [79]			HAWKESBURY SANDSTONE 0.93 m: DB 1.00 m: HB 1.03 m: P, 10°, RF, PR, SN Fe 1.13 m: P, 15°, RF, PR, CN 1.24 m: DB
					61.0		from 1.50 to 2.00 m: grading from brown to pale grey					
	2		Is(50) D=0.51 MPa Is(50) A=0.74 MPa Is(50) D=0.59 MPa Is(50) A=0.95 MPa		2.0		from 2.30 to 3.00 m: becoming red and brown		100 (100) [96]			2.00 m: P, 5°, RF, PR, CN 2.05 m: P, 0°, RF, PR, VN Fe 2.08 m: P, 0°, RF, PR, SN Fe 2.30 m: P, 5°, RF, PR, CN 2.36 m: P, 0°, RF, PR, SN Fe
					60.0							2.67 m: J, 50°, SM, CU, calcite, healed, 130 mm persistence 2.76 m: J, 75°, PR, healed
					3.0							3.00 m: HB
			Is(50) D=0.75 MPa Is(50) A=0.8 MPa		3.32		SANDSTONE: medium to coarse grained, pale grey, thinly cross-bedded at 5-20°.	SW to FR	100 (100) [100]			3.57 m: J, 60°, UN, healed
	3				4.0							3.95 m: HB 4.00 m: HB 4.06 m: DB
					58.0							
			Is(50) D=1.24 MPa Is(50) A=1.28 MPa		4.83				100 (98) [98]			5.00 m: HB 5.04 m: P, RF, IR, CN X
	4		Is(50) D=1.66 MPa Is(50) A=2.07 MPa		5.0							5.37 m: DB at 5.50 m: 100% water loss 5.50 m: P, 5°, RF, IR, CN, 35 mm aperture
					57.0							5.80-5.82 m: EW, 20°, RF, IR, clay, 20 mm thick
			Is(50) A=1.91 MPa Is(50) D=1.54 MPa		6.0				100 (100) [98]			5.86 m: DB 6.00 m: HB 6.05 m: DB
					56.0							
	5		C: 7.32-7.52 m UCS=39 MPa		7.0							at 7.00 m: 0% water loss 7.00 m: HB 7.06-7.09 m: J, 25-30°, RF, PR, CN, x 2
			Is(50) D=1.85 MPa		7.81			FR	100			
					8.0							

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HQ3	6		Is(50) A=1.76 MPa	54.0			SANDSTONE: medium to coarse grained, pale grey, thinly cross-bedded at 5-20°.	FR	(100) [100]			7.98 m: HB at 8.00 m: 80% water loss 8.00 m: HB 8.26 m: P, 0°, RF, PR, CN
			Is(50) D=1.74 MPa Is(50) A=1.57 MPa	9.0 53.0					100 (100) [100]			8.96 m: P, 5°, RF, PR, CT clay, 5 mm thick 9.00 m: HB 9.06 m: P, 15°, RF, PR, CN 9.24 m: P, 10°, RF, PR, CN
	7		Is(50) D=2.16 MPa Is(50) A=2.54 MPa	10.0 52.0			from 9.77 to 9.95 m: trace medium, angular, shale and quartz gravels					9.65 m: DB 9.78 m: HB 10.00 m: HB
			Is(50) D=1.94 MPa Is(50) A=1.4 MPa	11.0 51.0					100 (100) [100]			11.00 m: HB
	8		Is(50) D=2.45 MPa Is(50) A=1.53 MPa	12.0 50.0								11.58 m: P, 20°, RF, PR, CN 11.83 m: DB 12.00 m: HB
			Is(50) D=1.88 MPa Is(50) A=1.81 MPa	13.0 49.0					100 (100) [100]			12.15 m: J, 75°, PR, healed 12.16 m: P, 5°, RF, IR, CT clay 12.48 m: P, 5°, RF, UN, CN
	9		C: 13.11-13.31 m UCS=30 MPa	13.0 49.0								13.00 m: HB
			Is(50) D=1.92 MPa	14.0 48.0					100 (100) [100]			13.95-14.00 m: HB 14.35 m: DB
	10		Is(50) A=1.99 MPa Is(50) D=1.86 MPa Is(50) A=2.36 MPa	15.0 47.0			from 14.46: trace dark grey laminations from 14.52 to 14.80 m: disturbed bedding		100 (100) [100]			15.00 m: HB 15.15 m: DB
			Is(50) D=1.77 MPa Is(50) A=1.93 MPa	16.0 46.0								

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HQ3	11				46.0		SANDSTONE: medium to coarse grained, pale grey, thinly cross-bedded at 5-20°.	FR				15.96 m: HB 16.00 m: HB
	16.87											
			Is(50) D=1.64 MPa Is(50) A=2.16 MPa		17.0				100 (100) [100]			16.75 m: P, 15°, RF, PR, VN clay 16.87 m: DB 17.00 m: HB
					45.0							
	12											
			Is(50) D=2.02 MPa Is(50) A=2.2 MPa		18.0							17.96 m: HB 18.00 m: HB
			Is(50) D=1.25 MPa Is(50) A=1.47 MPa		44.0				100 (100) [100]			18.18 m: P, 15°, RF, PR, clay
	18.39											
					19.0							19.00 m: HB
HQ3	13						from 22.18 to 22.38 m: disturbed laminations					
			C: 19.12-19.32 m Is(50) D=2.09 MPa UCS=40 MPa Is(50) A=1.77 MPa		43.0							
					20.0							
			Is(50) D=2.37 MPa Is(50) A=1.64 MPa		42.0				100 (100) [100]			20.00 m: HB 20.15 m: P, 35°, RF, PR, CN
	14											
			Is(50) D=2.36 MPa Is(50) A=2.23 MPa		21.0							20.95 m: P, 10°, RF, PR, VN clay 20.98 m: HB 21.00 m: HB 21.10 m: HB
					41.0				100 (100) [100]			
	21.37											
					22.0							
HQ3	15						from 22.18 to 22.38 m: disturbed laminations					
			C: 22.00-22.44 m UCS=24 MPa		40.0							
			C: 22.44-22.81 m									
HQ3	16						from 22.18 to 22.38 m: disturbed laminations					
			Is(50) D=1.4 MPa Is(50) A=2.05 MPa		23.0				100 (100) [100]			23.00 m: HB
			Is(50) D=1.73 MPa Is(50) A=2.16 MPa		39.0							23.44 m: P, 5°, RF, PR, VN clay
	22.86											

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CORE PHOTO REPORT

ID NO. : WSM_BH14

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Sheet 1 of 3



Core Tray - 0.70 to 4.00 m



Core Tray - 4.00 to 8.00 m



Core Tray - 8.00 to 12.00 m

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CORE PHOTO REPORT

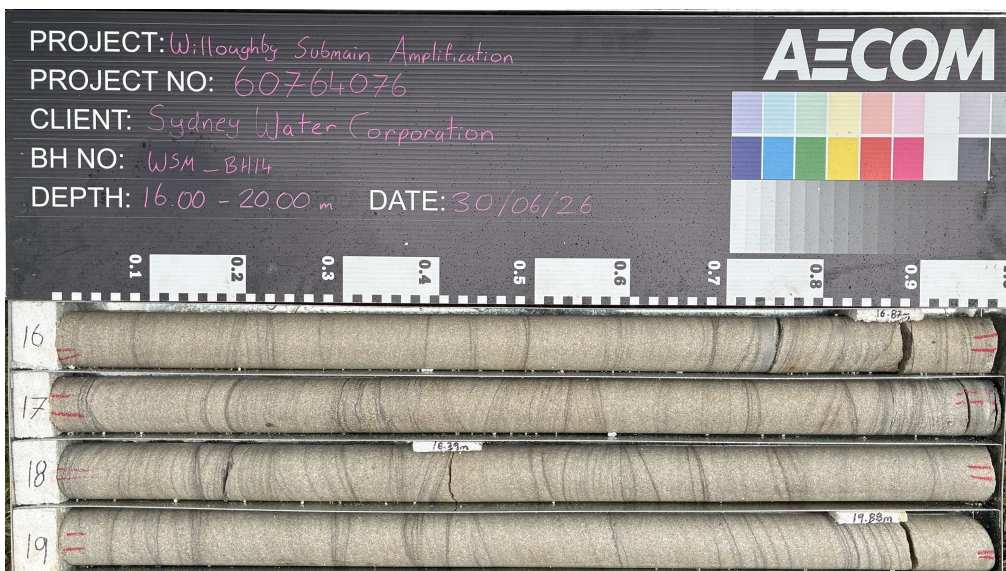
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Core Tray - 12.00 to 16.00 m



Core Tray - 16.00 to 20.00 m



Core Tray - 20.00 to 24.00 m

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CORE PHOTO REPORT

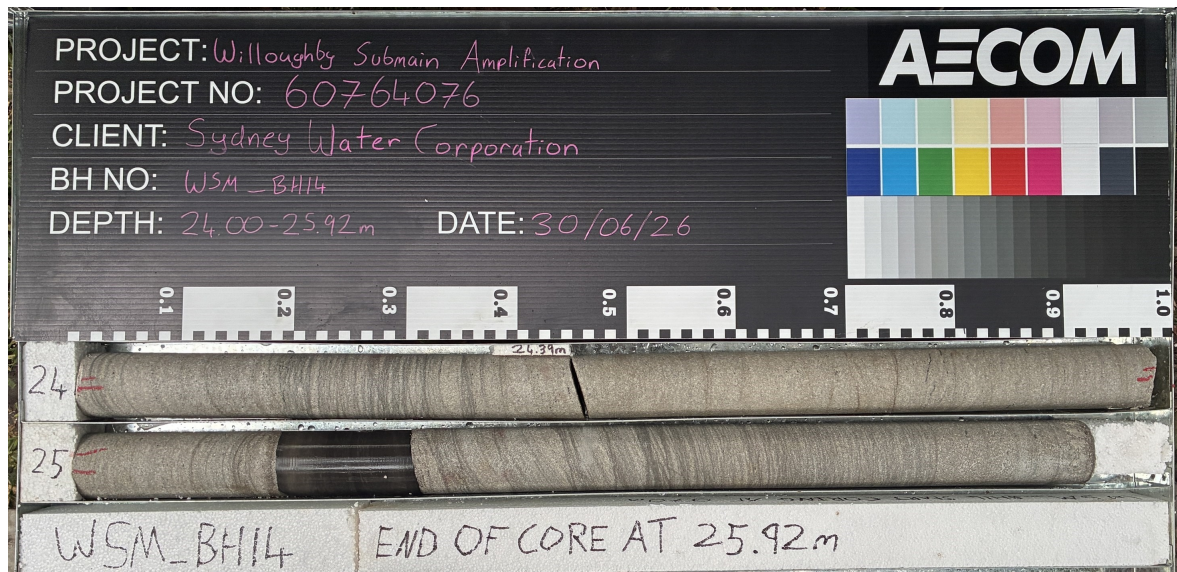
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